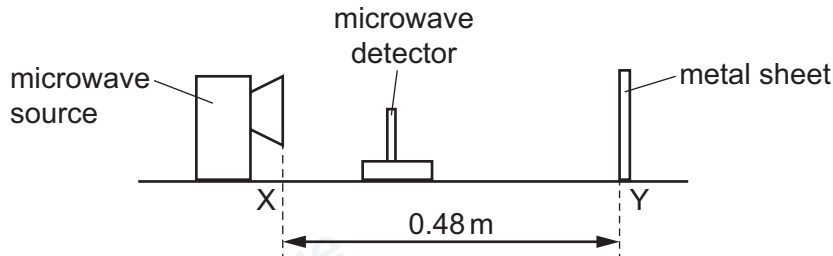


30 A microwave source is placed on a horizontal surface at position X.

A shiny metal sheet is placed at position Y that is a distance of 0.48 m from X.



The microwaves are reflected back from the metal sheet towards the source.

A microwave detector is moved from X to Y and all positions of minimum microwave intensity are recorded.

Intensity minima were detected at X and Y and also at 15 positions in between X and Y.

The speed of the microwaves is $3.0 \times 10^8 \text{ m s}^{-1}$.

What is the frequency of the microwaves emitted by the source?

- A** $4.7 \times 10^9 \text{ Hz}$ **B** $5.0 \times 10^9 \text{ Hz}$ **C** $9.4 \times 10^9 \text{ Hz}$ **D** $1.0 \times 10^{10} \text{ Hz}$